Optimality of the endpoint Orlicz loss: every dense Walsh subsystem contains an affine spike

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Abstract

Burstein proves that every uniformly bounded orthogonal system of size n has a subsystem of cardinality at least $n/[e(\log n)^{\alpha+1}]$ whose $L^2 \log^\alpha L$ synthesis norm is at most $C_\alpha(\log\log n)^{\alpha/2}$, and asks whether the iterated logarithm is optimal. We prove that it is. For every sufficiently large $N = 2^d$, take all characters of $\mathbb{F}_2^d$. Every subset of the target cardinality contains an affine frequency subspace of size $m \asymp_\alpha \log N/\log\log N$. The normalized character sum on that subspace is exactly $\sqrt{m}$ on a set of measure $1/m$ and zero elsewhere, so its Luxemburg norm is at least $c_\alpha(\log m)^{\alpha/2} \asymp c_\alpha(\log\log N)^{\alpha/2}$. The affine-subspace assertion follows from an elementary cube count and a count of linearly dependent direction tuples. Thus the source upper bound has the optimal asymptotic order in the worst case.

1 Source question

The source is Will Burstein, Λ_p *Style Bounds in Orlicz Spaces Close to* L^2, arXiv:2505.00155v2 [1]. Its Theorem 2.7 proves the upper bound discussed below. On page 13, Remark 2.11 says that the optimality of equation (2.4) remains open. The relevant source passage is reproduced here.

$$\|f\|_\Phi/\|a\|_2 \leq K(\alpha)\cdot \log^{\frac{\alpha}{2}}(u_0 + S^2 \log n),$$

which completes the proof. □

2.3. **Probabilistic Sharpness Result.** The result of this section uses some ideas similar to those of the probabilistic sharpness result of [2]. As before, constants can vary line by line.

Theorem 2.10. *Fix $\rho > 0$ and $\alpha > 0$ and set $n := \frac{m}{2\rho} m^m$. Consider the probability space $[0,1]$ with Lebesgue measure. For $k \in [n]$, we set*

$$\varphi_k := \exp(-2\pi i k(\cdot)),$$

the orthonormal characters on $[0,1]$. Let $\{\zeta_k\}_{k=1}^n$ be $\{0,1\}$-valued and i.i.d. with mean $\delta = \frac{1}{\log^\rho(n)}$. Let J_ω be the random set,

$$J_\omega := \{k \in [n] : \zeta_k(\omega) = 1\}.$$

Let $\Phi(u) = u^2 \log^\alpha(u)$ for $u \geq u_0$, and $\Phi(u) = c(\alpha)u^2$ otherwise. $u_0 \geq e$ and $c(\alpha)$ are constants, depending on α, chosen so that Φ is a Young function. Then,

$$\mathbb{E}_J \left(\sup_{\|a\|_2 \leq 1} \left\| \sum_{k \in J} a_k \varphi_k \right\|_\Phi \right) \geq K(\alpha, \rho) \log^{\frac{\alpha}{2}}(\log n). \tag{2.13}$$

Remark 2.11. We state our main result, Theorem 2.7, in terms of a probability over subsets of $[n]$ following a Bernoulli distribution. We could have, just as easily, stated our main result in terms of an expectation over subsets of $[n]$ as in equation 2.13. The latter form, using the expectation, is the formulation of Theorem 1.5 of [12] (we note that the main technical machinery of Theorem 1.5 is Lemma 4.3, of [12], which uses similar techniques to Theorem 2.7). Hence, we can rule out the common randomized technique, subset selection of $[n]$ by a Bernoulli distribution on $[n]$'s indices, to show the existence of a subset which sharpens the bound of equation 2.4. The question of whether the bound of equation 2.4 is optimal remains open, but equation 2.13 gives evidence that the bound is indeed optimal.

Fix $\alpha > 0$. The source uses a Young function Φ for which, with constants $u_0 \geq e$ and $c(\alpha) > 0$ fixed once α is fixed,

$$\Phi(u) = u^2 \log^\alpha u \quad (u \geq u_0), \qquad \Phi(u) = c(\alpha)u^2 \quad (0 \leq u < u_0).$$

The corresponding Luxemburg norm on a probability space is

$$\|f\|_\Phi = \inf\{\lambda > 0 : \mathbb{E}\Phi(|f|/\lambda) \leq 1\}.$$

For an orthogonal family $(\varphi_\xi)_{\xi \in I}$, write

$$\mathcal{N}_\Phi(I) = \sup_{\|a\|_2 \leq 1} \left\| \sum_{\xi \in I} a_\xi \varphi_\xi \right\|_\Phi.$$

The source proves that every bounded orthogonal system of size n has, with positive probability and hence in particular deterministically, a subset I with

$$|I| \geq \frac{n}{e(\log n)^{\alpha+1}} \quad \text{and} \quad \mathcal{N}_\Phi(I) \leq C_\alpha (\log\log n)^{\alpha/2}.$$

It asks whether the factor $(\log\log n)^{\alpha/2}$ can be improved.

2 Proof intuition

The source lower bound finds a consecutive Fourier block inside a random subset. That mechanism cannot control a deterministic subset, which may avoid all long intervals. The replacement is to use all binary characters and search for an affine subspace of frequencies. An affine r-flat is an ideal bad configuration: the sum of its $m = 2^r$ characters is supported on the annihilator of its direction space, an event of probability exactly $1/m$, and has normalized height exactly $\sqrt{m}$ there.

It remains to force such a flat inside *every* target-size subset. If $A \subset \mathbb{F}_2^d$ has density δ, the normalized number of ordered r-dimensional cubes in A is at least δ^{2^r}. This follows by iterating Cauchy–Schwarz (in the particularly simple form of Jensen's inequality for opposite cube faces). Degenerate cubes, whose directions are linearly dependent, account for fewer than $2^r/2^d$ of all direction tuples. Hence A contains a genuine affine r-flat as soon as $\delta^{2^r} 2^d > 2^r$.

At the source density $\delta \asymp (\log N)^{-(\alpha+1)}$, this condition permits $2^r \asymp_\alpha \log N / \log\log N$. Since the Orlicz norm of the resulting spike grows as $(\log 2^r)^{\alpha/2}$, the desired endpoint loss follows.

3 The deterministic lower theorem

Let $G_d = \mathbb{F}_2^d$, let $N = |G_d| = 2^d$, and give G_d uniform probability measure. For $\xi, x \in G_d$, define

$$\chi_\xi(x) = (-1)^{\xi \cdot x},$$

where the dot product is over $\mathbb{F}_2$. The N functions $(\chi_\xi)_{\xi \in G_d}$ are orthonormal and have supremum norm one.

Theorem 1 (Worst-case optimality)**.** *Fix $\alpha > 0$ and let Φ be the Young function above. There are $c_\alpha > 0$ and $d_0(\alpha)$ such that for every $d \geq d_0(\alpha)$, the character system $(\chi_\xi)_{\xi \in \mathbb{F}_2^d}$ has the following property. Every $I \subseteq \mathbb{F}_2^d$ satisfying*

$$|I| \geq \frac{N}{e(\log N)^{\alpha+1}}, \qquad N = 2^d,$$

obeys

$$\mathcal{N}_\Phi(I) \geq c_\alpha (\log\log N)^{\alpha/2}.$$

Consequently the factor $(\log\log n)^{\alpha/2}$ in the source upper bound is optimal, up to constants depending only on α, in the worst-case best-subsystem problem.

We first prove the combinatorial statement which replaces random block selection.

Lemma 2 (Dense sets contain affine cubes)**.** *Let $A \subseteq \mathbb{F}_2^d$ have density $\delta = |A|/2^d$. If an integer $r \geq 1$ satisfies*

$$\delta^{2^r} 2^d > 2^r,$$

then A contains an affine subspace of dimension r.

Proof. Put $G = \mathbb{F}_2^d$, $N = |G|$, and $f = \mathbf{1}_A$. For $s \geq 0$, let

$$C_s = \mathbb{E}_{x, h_1, \ldots, h_s \in G} \prod_{\omega \in \{0,1\}^s} f\left(x + \sum_{j=1}^{s} \omega_j h_j\right).$$

Thus C_s is the normalized number of ordered s-cubes contained in A, with degenerate cubes included, and $C_0 = \delta$.

For fixed $h_1, \ldots, h_s$, set

$$g(x) = \prod_{\omega \in \{0,1\}^s} f\left(x + \sum_{j=1}^{s} \omega_j h_j\right).$$

Splitting an $(s+1)$-cube into its two opposite s-faces and averaging first in its final direction gives

$$C_{s+1} = \mathbb{E}_{h_1,\ldots,h_s} \mathbb{E}_{x,t} g(x) g(x+t) = \mathbb{E}_{h_1,\ldots,h_s} (\mathbb{E}_x g(x))^2 \geq C_s^2.$$

The last step is Jensen's inequality. Induction yields

$$C_r \geq \delta^{2^r}.$$

Therefore the number of ordered tuples $(x, h_1, \ldots, h_r)$ defining a cube in A is at least $\delta^{2^r} N^{r+1}$.

We now bound degenerate tuples. Choosing the directions in order, if $h_1, \ldots, h_r$ are linearly dependent, then for some first index j one has $h_j \in \operatorname{span}(h_1, \ldots, h_{j-1})$. For each j, there are at most

$$N^{j-1} 2^{j-1} N^{r-j} = 2^{j-1} N^{r-1}$$

such ordered direction tuples. Hence fewer than $(2^r - 1) N^{r-1}$ direction tuples are dependent, and after choosing x fewer than $(2^r - 1) N^r$ ordered cubes are degenerate.

The assumed inequality makes the total number of cubes strictly larger than the number of degenerate tuples. Thus one cube has linearly independent directions. Its 2^r vertices form an affine r-dimensional subspace contained in A. $\square$

Lemma 3 (The target density forces a large flat)**.** *Put $\rho = \alpha + 1$. For all sufficiently large $N = 2^d$, every $A \subseteq \mathbb{F}_2^d$ with*

$$|A|/N \geq \frac{1}{e(\log N)^{\rho}}$$

contains an affine subspace of cardinality $m = 2^r$ satisfying

$$\frac{\log N}{16\rho \log\log N} < m \leq \frac{\log N}{8\rho \log\log N}.$$

Proof. Let $m = 2^r$ be the largest power of two not exceeding

$$M = \frac{\log N}{8\rho \log\log N}.$$

For large N, $m > M/2$, giving the displayed bounds. If $\delta = |A|/N$, then

$$\begin{aligned}\log\left(\frac{\delta^m N}{m}\right) &\geq \log N - m - \rho m \log\log N - \log m \\ &\geq \log N - M(1 + \rho \log\log N) - \log M.\end{aligned}$$

For all sufficiently large N, the last expression is positive (indeed it is at least $\frac{3}{4}\log N - \log M$). Thus $\delta^m N > m$, and Lemma 2 applies. $\square$

Lemma 4 (An affine character flat is an Orlicz spike)**.** *There are $b_\alpha > 0$ and $m_0(\alpha)$ such that the following holds. If $V = \xi_0 + H \subseteq \mathbb{F}_2^d$ is an affine subspace of cardinality $m \geq m_0(\alpha)$, then*

$$\left\| m^{-1/2} \sum_{\xi \in V} \chi_\xi \right\|_{\Phi} \geq b_\alpha (\log m)^{\alpha/2}.$$

Proof. Let

$$H^\perp = \{x \in \mathbb{F}_2^d : h \cdot x = 0 \text{ for every } h \in H\}.$$

Character orthogonality on the group H gives

$$\left| m^{-1/2} \sum_{\xi \in V} \chi_\xi(x) \right| = m^{-1/2} \left| \sum_{h \in H} \chi_h(x) \right| = \begin{cases} \sqrt{m}, & x \in H^\perp, \\ 0, & x \notin H^\perp. \end{cases}$$

Since $|H^\perp| = N/m$, the nonzero event has probability $1/m$.

Set $\lambda = b_\alpha (\log m)^{\alpha/2}$. For large m, $\sqrt{m}/\lambda \geq u_0$ and

$$\log(\sqrt{m}/\lambda) \geq \tfrac{1}{4} \log m.$$

Consequently

$$\mathbb{E}\Phi \left(\frac{m^{-1/2} |\sum_{\xi \in V} \chi_\xi|}{\lambda} \right) = \frac{1}{\lambda^2} \log^\alpha(\sqrt{m}/\lambda) \geq \frac{4^{-\alpha}}{b_\alpha^2}.$$

Choose, for example, $b_\alpha = 2^{-\alpha-1}$. The last quantity is at least $4 > 1$. By the definition of the Luxemburg norm, the norm is at least λ. □

Proof of Theorem 1. Let I have the stated cardinality. By Lemma 3, I contains an affine subspace V of cardinality m with

$$m \asymp_\alpha \frac{\log N}{\log \log N}.$$

Put coefficients $a_\xi = m^{-1/2}$ on V and zero on $I \setminus V$. They have ℓ^2 norm one, so Lemma 4 gives

$$\mathcal{N}_\Phi(I) \geq b_\alpha (\log m)^{\alpha/2}.$$

For sufficiently large N,

$$\log m \geq \log \log N - \log \log \log N - \log(16\rho) \geq \tfrac{1}{2} \log \log N.$$

Absorbing the factor $2^{-\alpha/2}$ into the constant proves the lower bound. The source upper bound holds for every uniformly bounded orthogonal system, so the two estimates establish optimality in worst-case asymptotic order. □

4 Verification and novelty notes

The proof has no unproved dependency and no computational step. Its two quantitative thresholds can be audited directly:

- the cube lower bound counts all ordered cubes, while the explicit $(2^r - 1)N^{r-1}$ estimate counts every dependent direction tuple;
- at density $e^{-1}(\log N)^{-\rho}$, taking $2^r \leq \log N / [8\rho \log \log N]$ leaves a positive exponential margin in the condition $\delta^{2^r} N > 2^r$;
- the affine character sum is an exact two-valued function, so the Luxemburg modular calculation has no tail estimate or asymptotic approximation other than choosing N sufficiently large.

A bounded novelty check on August 11, 2026 searched the run registry and solution/attempt indexes for arXiv:2505.00155, endpoint Orlicz optimality, Walsh systems, affine subspaces, and cube-count terms. External searches covered the exact title and arXiv id, combinations of "bounded orthogonal systems", "Orlicz", "Walsh", "affine subspace", and "Gowers norm", and recent papers returned for affine-subspace statistics. The search found the source paper and unrelated/background work, but no later paper claiming to answer Remark 2.11 and no matching deterministic lower theorem. Novelty therefore remains subject to expert literature review.

5 Limitations and human-review recommendation

The theorem is stated on the infinite dimension subsequence $N = 2^d$, which is sufficient to prove that the source factor cannot be uniformly replaced by a smaller asymptotic order. Constants and the threshold d_0 depend on α, exactly as in the source upper bound. No assertion is made about the sharp numerical constant.

This is recommended for expert review as a likely valid full solution of the source optimality question. The highest-value checks are: (i) the $C_{s+1} \geq C_s^2$ cube-count recursion, (ii) the subtraction of degenerate direction tuples, (iii) the scale choice in Lemma 3, and (iv) the precise match between the source's best-subsystem quantifiers and the universal statement for the constructed character system.

References

[1] W. Burstein, Λ_p *Style Bounds in Orlicz Spaces Close to* L^2, arXiv:2505.00155v2, 2025.